Sthitadhi Maiti

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EDUCATION

Postdoctoral Research Associate in Computational Biophysics,
Department of Physics, University of Genoa, Genova, Italy

PhD in Chemistry, May 2024
School of Molecular Sciences, Arizona State University, Tempe AZ (CGPA: 3.76/4)

Master of Science in Chemistry, May 2016
Indian Institute of Technology Guwahati, India

Bachelor of Science in Chemistry, May 2014
Ramakrishna Mission Vidyamandira, Belur Math, India

TECHNICAL SKILLS & INTERESTS

Technical Skills: *Chemistry/Biophysics/Statistics*- Research | Computational Chemistry | Biochemistry | Classical Molecular Dynamics | Multiscale Simulations | Enhanced Sampling | Umbrella sampling | Free Energy Calculations | FEP | Statistical Data Analysis | Drug design | Structure & Ligand Based Virtual Screening | Docking | Shape Screening | R-group Enumeration.

Programming/Software- Wolfram Mathematica | GROMACS | PLUMED | Martini | VMD | PyMOL | Chimera | Schrodinger Maestro | LiveDesign | Bash scripting | Python | C | AWK | Linux | Git | HPC | Rosetta | PyRosetta | Wavefunction Spartan | Modeller | AutoDock Vina | Pandas | Matplotlib | Machine Learning | RDKit | scikit-learn | PyTorch | Tensorflow | R | Blender | LaTeX | ChemDraw | SQL | Excel

Experimental- Organic/Organometallic/Material Synthesis | Extraction and Purification | IR | UV-Vis | Fluorescence | NMR | GC-MS | Powder XRD | MALDI

Communication Skills: Teaching assistant in undergraduate chemistry labs (Gen Chem 113, 116) for six semesters.

Interests: Reading popular science articles | Graphic Designing on Photoshop, Illustrator & Blender | Photography | Hiking

RESEARCH EXPERIENCE

Postdoctoral Research Associate at Dr. Davide Bochicchio, and Prof. Giulia Rossi's NanoBioComp Lab, UniGe, Genoa, Italy

- Develop and validate a novel **collective variable (CV)** in **PLUMED** and Python to quantify alpha-helical and beta-sheet content of amyloid beta-42 in CG simulations, with flexibility for adaptation to other foldable CG protein models.
- Examine how **foldable coarse-grained models** of amyloid-β and other **small proteins** interact with **cell membranes** of diverse lipid compositions, with particular emphasis on binding mechanisms and membrane pore formation.

Research Assistant at Dr. Matthias Heyden's Computational Chemistry Lab, ASU (08/2018 – 05/2024)

- Study the impact of **protein-water interactions** and empirical scaling parameters on the thermodynamics of protein solvation with explicit solvent **classical MD simulations**, using K-18 domain of disordered Tau protein as the model system.
  ---- *J. Phys. Chem. B.* **2023**, 127, 33, 7220–7230.
- Generate and analyze 3D maps of local contributions to the solvation free energy of **intrinsically disordered proteins** (programming in **Mathematica, and C**).
- Explore the feasibility to design peptide-based ligands with high binding affinity to α-MoRF (alpha-Molecular Recognition Feature) domains based on binding-induced folding mechanisms.
- Develop an automated simulation protocol for the calculation of peptide folding free energy profiles using simulation package of **GROMACS**, **PLUMED,** and **enhanced sampling** technique of **Umbrella Sampling.**
- Find **binding free energies** of antiapoptotic protein Mcl-1 to PUMA and other α-MoRF BH3-only domains using **coarse grained umbrella sampling** in SIRAH forcefield, with an aim to correlate them with the folding free energy penalties of the BH3 domains. **----** *Protein Dynamics*; Academic Press, **2026**, 215–240, ISBN 9780443300424.
- Two **collaborative research works** with professors in Italy and at ASU resulting in two publications.

Research Assistant at Dr. Ryan Trovitch's Organometallic Synthesis Lab, ASU (07/2016 – 08/2018)

- Designed, and performed, organometallic synthesis, and analyses such as IR, and NMR.
- Optimized separation/purification techniques -- column chromatography and crystallization of compounds.

- Performed **organic synthesis** of a modified Gd-DOTA MRI contrast agent to be used selectively in cells under hypoxic conditions. Used MALDI-TOF, and UV-Vis to analyze it.

Research Intern at Dr. Sebastian C. Peter's Solid State & Inorganic Chemistry Lab, JNCASR, India (05/2015 – 07/2015)

- Developed, synthesized, and characterized (with powder XRD & SC-XRD) heterogeneous phosphomolybdates, and phosphotungstate catalysts for small molecule oxidation. Published in Inorganic Chemistry in 2018.

Research Intern (M.Sc.) at Dr. Gopal Das' Supramolecular Chemistry Lab, IIT Guwahati, India (11/2015 – 05/2016)

- Developed, synthesized, and characterized (with UV-vis, IR and NMR) a colorimetric and fluorescence "Turn-On"sensor for the detection of Hg(II) and Cu(II) ions.

EXTRACURRICULAR EXPERIENCE AND AWARDS

- Judged high school students' science experiments as a Grand Award Judge (Chemistry) for Intel Science and Engineering Fair 2019, May 14-15, Phoenix, AZ.
- Manage the Amateur Radio Club at ASU (W7ASU) as one of its current officers and designed the current logo of W7ASU.
- Attended weekly meetings of Machine Learning Club, Sun Devil Data Science, Linux Users Group, Code Devils and SoDA at ASU.
- Awarded the JBNSTS (Jagadish Bose National Science Talent Search) 2011 Senior Scholarship & DST Inspire Fellowship, Govt. Of India for the period of 2011-2016.

ACADEMIC CONFERENCES AND WORKSHOPS

- Attended the 15th European Biophysics Congress (EBSA) in Rome, Italy, in 2025 and presented a poster titled "A Collective Variable for Secondary Structure Assessment in a Foldable Coarse-Grained Model of Amyloid-β42."
- Attended and presented a poster at the ACS Fall 2019 Conference in San Diego on "Solvation Thermodynamics of Intrinsically Disordered Proteins (IDPs)."
- Presented poster at the Les Houches Protein Dynamics Workshop at Aussois, France, in 2022 on Free Energy Surface Contribution for IDPs in Atomistic Simulations.
- Research poster presentation at Biophysical Society 2023 Feb 18-22 in San Diego.
- Poster presentation at ACS Spring 2023 in Indianapolis, and at ACS Fall 2023, San Francisco.
- **Supervised Machine Learning**: Regression and Classification. https://coursera.org/share/2d7c16350b79f5eccbb30f49eb32b16e
- Schrodinger Online Courses: 1. **Introduction to Molecular Modeling in Drug Discovery**. (Completed 11/15/2023)
  https://courses.schrodinger.com/certificates/jontoiwo1u
  2. **High-Throughput Virtual Screening for Hit Finding and Evaluation**. (Completed 02/27/2024)
  https://courses.schrodinger.com/certificates/fr06qf4f3q
  Coursera - 1. **Supervised Machine Learning**: Regression and Classification.
  https://coursera.org/share/2d7c16350b79f5eccbb30f49eb32b16e
  Uni. Genova - 1. **Machine Learning Methods for Physics (offline course)**.
  https://corsi.unige.it/off.f/2025/ins/87930

PUBLICATIONS

(6) **Maiti, S.**; Heyden, M. Model-Dependent Solvation of the K-18 domain of the Intrinsically Disordered Protein Tau. *J. Phys. Chem. B.* **2023**, 127, 33, 7220–7230.

(5) Heyden, M.; **Maiti, S.** Binding-Induced Folding of Intrinsically Disordered Peptides. *Protein Dynamics*; Academic Press, **2026**, 215–240, ISBN 9780443300424.

(4) Nibali, V. C.; **Maiti, S.**; Saija, F.; Heyden, M.; Cassone, G. Electric-field induced entropic effects in liquid water. *J. Chem. Phys.* **2023**, 158, 184501.

(3) Sauer, M.; Mondal, S.; Neff, B.; **Maiti, S**.; Heyden, M. Fast Sampling of Protein Conformational Dynamics. *Sci. Adv.* **2026**, **12**, eaea4617. https://arxiv.org/abs/2411.08154.

(2) Sauer, M.; Colburn, T.; **Maiti, S**.; Heyden, M.; Matyushov, D. Linear and Nonlinear Dielectric Response of Intrinsically Disordered Proteins. *J. Phys. Chem. Lett.* **2024**, 15, 20, 5420–5427.

(1) Roy, S.; Vemuri, V.; **Maiti, S.**; Manoj, S. K.; Subbarao, U.; Peter, S. C. Two Keggin-based isostructural POMOF hybrids: synthesis, crystal structure, and catalytic properties. *Inorg. Chem.* **2018**, 57, 19, 12078–12092.